SolveThe17 2.0

Track 4 · SDG 11

CityFix

A cleaner city, faster services, and waste that becomes value.

Unlocking Egypt's Waste Economy

Report it. We follow it.



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Lots of waste, value lost, and workers who go unseen

~20M

tonnes of municipal solid waste a year

Theme fact EEAA

80%

up to, recovered by the Zabbaleen

Theme fact SolveThe17 brief

The gap

No structured framework connects households, collectors and the city, so the recovery stays informal and unmeasured.

Figures are quoted from the SolveThe17 2.0 Track 4 theme brief; we have not measured them ourselves.

How might we establish a structured circular economy framework in Egypt that transforms waste streams into economic value while elevating informal sector workers and improving urban environmental health?

Our answer: one platform that connects households, collectors and the city, in three parts.

1 A structured framework

Published price list and pickup requests	Built
Collector mode: nearby jobs and weighing	Built
Smart bins and hub scale	Simulated

2 Economic value

Pay per kg from a price list (demo prices)	Built
Impact ledger: kg and CO_2 avoided	Built
Measured in a field pilot	Planned

3 Workers and urban health

Collectors see jobs and earnings openly	Built
Street reports followed until fixed	Built
Collector verification and training	Planned

Built · Simulated · Designed · Planned — Status today, from the repo

The solution

One platform for city life: six pillars, not only recycling

FixFirst

Report problems with photo, voice and GPS; follow each one until it is fixed

Built

CityServe

Book city services from the same app

Built

CityCycle

Recycling pickups, collector mode, material prices, impact, smart bins

Built Simulated

CityOps

Work orders and department assignment for the district

Built

CityPulse

Live map: incidents, smart bins and dumping hotspots

Built

CityBrain

AI triage, fill forecast and route preview. AI only suggests

Built To verify

Built Simulated Designed Planned Status today, from the repo

Channels

One citizen, many channels: app, web, sorting guide, collector mode, smart bins

Citizen

- Built — Mobile app (Expo)
- Built — Citizen website
- Built — Photo and voice note (typing fallback)
- Built — Picture-based sorting guide

One backend, one database

- FastAPI · one API contract
- PostgreSQL / Supabase (SQLite locally)
- Device-key API for bins and scales
- AI suggestions, human confirms

City and field

- Built — Collector mode, same app
- Built — Command Center for staff
- Simulated — Smart bin over GSM
- Simulated — Hub scale at a collection point

The same report and the same record, whichever door you use.

Built · Simulated · Designed · Planned — Status today, from the repo

Who benefits

Three groups each get something they can see and measure

Citizens

Sort at home with a picture guide, book a pickup, see what material is worth, give it free or get paid, and watch reports get fixed

Informal collectors (Zabbaleen)

Nearby jobs, verified weights, a record of earnings, pay per kg from a published district price list

Districts

Bins emptied when full, shorter routes, visible dumping hotspots, tonnes diverted measured not estimated

From the repo vision. Benefits are what the platform records and shows; no savings or income figure is claimed before pilot data.

The demo story

Request, accept, weigh, confirm, impact, and a bin that calls for collection

1

Household requests a pickup

Verified in tests

`test_create_pickup_estimates_from_price_list`

2

Collector accepts the job

Verified in tests

`test_first_accept_wins_and_weighing_rules`

3

Collector weighs

Verified in tests

`test_full_flow`

4

Household confirms

Verified in tests

`test_full_flow`

5

Impact is recorded

Verified in tests

`test_impact_and_earnings`

And the smart bin calls for collection

A full or hot bin shows in the public bin list, and a citizen can report a full bin. Readings come from the simulator today.

Verified in tests

Simulated

Test names are in backend/tests/test_citycycle.py. Amounts are recorded only: no payment provider is integrated.

Report, follow, and book a service from one app

Citizen app

Pick the type; emergency numbers up front	Built
Photo, voice note, location	Built
AI suggests, the citizen confirms	Built
Follow it; "me too" on the map	Built



Home



Report



Follow it



CityServe

Screens: rendered from the team design files (mobile-Main, mobile-Report, mobile-Track, mobile-Services). The app runs on mock data by default.

CityCycle

CityCycle: households, price list, hubs and an impact ledger

Household

Picture sorting guide, pickup booking, free or paid, estimate before pickup

Built

District price list

The district sets prices. A material without a price gets no estimate. All prices shown are demo prices.

Built Demo values

Collection hubs

A scale records verified weight. The API route exists; the prototype keeps weighings in a local queue.

Built Simulated

Impact ledger

13.5 kg	diverted
51.5 EGP	value of weighed material
48 EGP	recorded as paid to households
5 / 1	pickups / job
0.0135 t	tonnes

Demo values Verified in tests

CO_2 is withheld until each material has a sourced factor. Amounts are recorded; no money moves.

Collector mode: clear jobs, verified weight, recorded earnings

What the collector gets

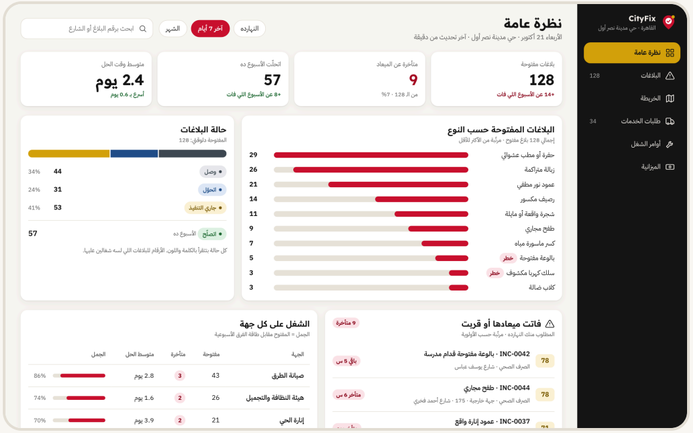
Nearby jobs on a map, approximate pin only	Verified in tests
Verified weights, weighings are idempotent	Verified in tests
Earnings record and payout list	Built
Works on weak networks: actions queue and sync later	Built
Picture-based, low-literacy interface	Designed

How trust is built

First accept wins; the second gets a 409	Verified in tests
The household confirms before it closes	Verified in tests
Collector applies; the district reviews	Built
Staff see the district, not the home address	Verified in tests
Training, insurance and ID checks	Planned

Collector mode is built in the mobile app (apps/mobile/src/app/collector), mock mode by default. Not yet field-tested with real collectors.

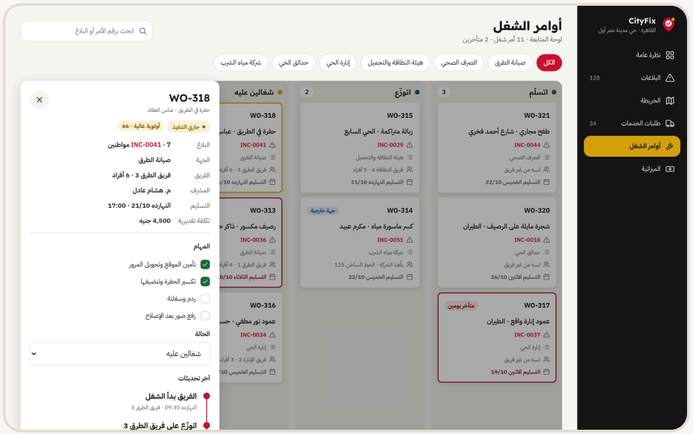
The Command Center: the district sees, ranks and decides



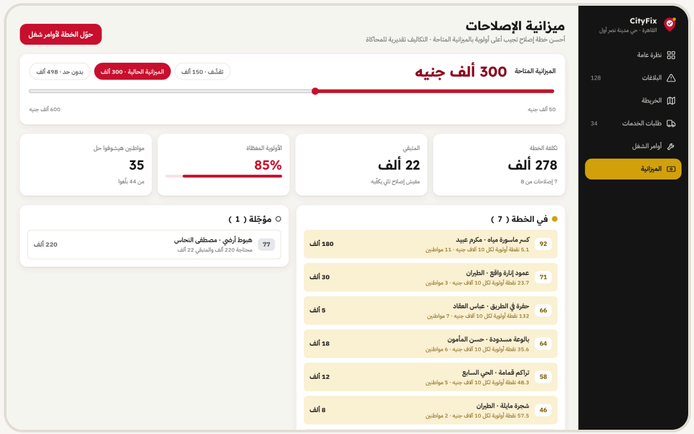
Overview



Incident detail



Work orders



Budget planner

Command Center

Incidents ranked; the priority score is explained	Built
Work orders and department assignment	Built
Budget planner picks the best repairs in budget	Built
Map: incidents, bins, dumping hotspots	Built
Bin fill forecast and route preview	Built

Screens from the design files. Forecast, route and bin screens exist in apps/admin-web (vitest-tested); not yet run on a deployed API.

CityBrain

CityBrain: AI that suggests, and a human who decides

1. Report with text, photo or voice
2. Gemini suggests category and severity
3. No Gemini: keyword fallback
4. A human confirms the suggestion
5. An explained priority score

Safeguards

AI only suggests; a human confirms	Built
Deterministic keyword fallback when Gemini is unavailable	Built
Photo analysis is off by default until the privacy gate is passed	Built
No accuracy claim: 33 synthetic cases, top-1 90.9%, top-3 100%, not a benchmark	Verified in tests

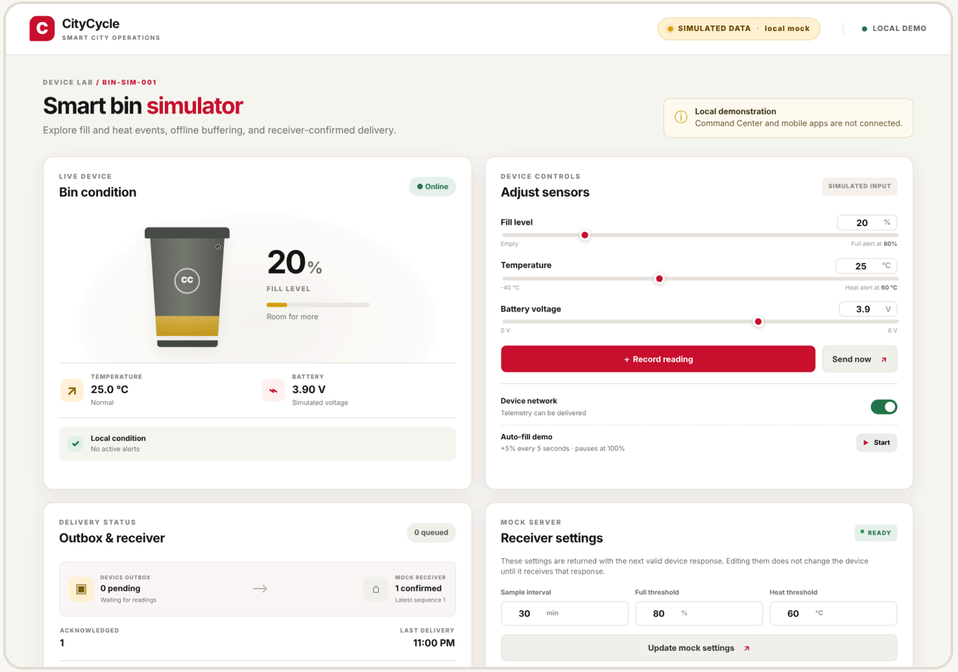
More in CityBrain

Linear fill forecast: 3+ readings over 1+ hour	Verified in tests
OR-Tools capacity-constrained route preview (pickups only; bins not stops yet)	Verified in tests
Dumping hotspots from repeat reports	Built
Gemini quality on Egyptian photos and voice	To verify

Smart bin and hub scale: designed and simulated, honestly labelled

The smart bin and hub scale

Ultrasonic fill-level sensor and a temperature probe	Designed
ESP32 deep-sleeps and wakes on a timer	Simulated
18650 cell with solar charging	Designed
SIM800L modem (2G only), LTE as the alternative	Designed
Device key plus sequence number; retries never duplicate	Verified in tests
Network drops: readings buffer and retry later	Simulated



PC simulator (hardware/simulator), local mock: real screenshot, not connected to the API

Simulated: no physical unit has been built or field-tested yet

Sources: hardware/README.md, hardware/schematics/smart-bin.svg, docs/field-brief/field-deployment-brief.md, backend/tests/test_bins.py.

Architecture

One backend, three clients, devices with a per-device key, AI that only suggests

Mobile app: citizen and collector — Built	Citizen website — Built	Command Center — Built

▼ ▲

One API contract · same endpoints and shapes for every client

PostgreSQL + PostGIS on Supabase; SQLite locally; Alembic migrations; row-level security — Built	Gemini triage with keyword fallback, linear forecast, OR-Tools route preview — Built	ESP32 smart bin and hub scale post to /iot with a device key — Simulated

Citizen OTP login; staff accounts	Per-device key; (binId, seq) is unique	Idempotency keys on writes	Redacted staff view; approximate pin

Source: docs/architecture.md and docs/diagrams/system-architecture.svg. Production deployment is not yet verified.

Technical rigor: what tests verify, and what is still open

430
backend tests passed (9 skipped)

66
simulator tests OK

33
synthetic AI smoke cases, not a benchmark

1 / folder
CI workflow per app folder

Verified in tests

First accept wins; weighings are idempotent	Verified in tests
Duplicate or conflicting readings are refused	Verified in tests
Staff see a district; collectors cannot locate a home	Verified in tests
Fill forecast and route partitioning	Verified in tests

Still to verify

Sensor accuracy in heat and dust, GSM and battery	To verify
Forecast accuracy on real bins	To verify
Gemini quality on consented Egyptian data	To verify
A live deployment, and OR-Tools in production	To verify

Counts recorded 2026-10-23 (docs/evidence.md); re-run before presenting. Reproduce: backend pytest, python -m unittest in hardware/simulator.

Impact: measured from platform data, not estimated

What we measure, and where the number comes from

kg diverted from landfill	Confirmed weighings (collector or hub scale)	Built
EGP earned by collectors	Weighings at the district price list, and payouts	Built
Pickup response time	Request to acceptance, and to completed weighing	To verify
Bins that never overflow	Share of time under the full threshold (threshold agreed with the district)	Simulated
Dumping hotspots cleared	Hotspots detected vs cleared, with time to clear	To verify
CO_2 avoided (secondary)	Withheld until each material has a sourced emission factor	To verify

Demo seed values today

13.5 kg	diverted (seed)
51.5 EGP	weighed value (seed)
48 EGP	recorded as paid (seed)
5	pickups (seed)
0.0135 t	tonnes (seed)

Demo values

No target, savings or income figure is promised before real pilot data.

Feasibility: local parts, and the field risks we already know

Parts in EGP, dated 2026-10-23 (partial)

ESP32	280 EGP
DS18B20 probe	65 EGP
Resistor kit	95 EGP
TP4056 charger	15 EGP
18650 cell	165 EGP
Cell holder	8 EGP
Schottky diode	1.25 EGP
5 V boost	25 EGP
P-MOSFET ×2	7 EGP
SIM800L GSM	300 EGP
Cable glands ×3	12 EGP

11 sourced lines = 973.25 EGP, one-time. Not a per-bin cost: probe, solar panel, enclosure and 14 more lines still need quotes.

Field readiness

Heat, dust and street washing: sealed box, breather vent, shaded sensor	Designed
Weak or no GSM: 32-reading buffer, external antenna, LTE option	Designed
Theft: security screws, nothing visible, tamper switch	Designed
Monthly cleaning; swap a pre-tested tray	Designed
Bench and field tests on a real unit	To verify

Sources: hardware/bom/bom.csv, docs/field-brief/field-deployment-brief.md. Design only: no unit has been built or placed on a street.

Evaluation framework

How each of the five criteria is answered, and where to see it

25	Local context and impact	Arabic-first, picture sorting, pay per kg for collectors, bins and hotspots for the district	Slides 2, 4–7, 16
25	Technical rigor and MVP	Working app, website and Command Center; tests; device-key telemetry; simulator	Slides 8–15
20	Innovation and interdisciplinarity	Software, AI, GIS, IoT hardware and a hub scale in one loop	Slides 5, 12–13
20	Economic and field feasibility	Local parts with dated EGP quotes, field risks and a pilot plan	Slides 17, 19
10	Technical communication and pitch	Architecture, docs, status tags and a 3-minute story	All slides

Points are the SolveThe17 2.0 evaluation framework (total 100). The slides show evidence; the scoring is the judges' call.

Next steps

Team, roadmap and a link to try it

Team

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Ziad Rabie
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Try it now

Report it. We follow it.

Roadmap

Today

Working app, website, Command Center; bins simulated

Pilot

One real bin and a district price list

Verify

Dated quotes, agreed thresholds, field tests

Deploy

Verified deployment, expand after open items close

Scan to open the citizen site and report something

Citizen web · city-fix-trojans.vercel.app

Command Center · city-fix-admin.vercel.app

API · city-fix-api-site.vercel.app

Links will be updated before the pitch. A live deployment is not yet verified in the repo.

Appendix A1: the API contract, grouped by who calls it

Citizen

`POST /otp` `POST /verify` `GET /me` `GET /me/impact` `POST /reports` `GET /incidents` `POST /incidents/{id}/follow` `POST /service-requests` `GET /map/incidents` `GET /notifications`

Recycling

`GET /recycling/materials` `GET /material-prices` `POST /recycling/pickups` `POST /recycling/pickups/{id}/confirm` `POST /recycling/pickups/{id}/cancel` `GET /recycling/summary`

Collector

`POST /collectors/apply` `GET /collector/jobs` `POST /collector/jobs/{id}/accept` `POST /collector/jobs/{id}/weighings` `POST /collector/jobs/{id}/complete` `GET /collector/earnings`

Staff and admin

`GET /dashboard` `PATCH /incidents/{id}` `POST /incidents/{id}/triage/confirm` `POST /work-orders` `POST /budget/optimize` `GET /admin/bins/forecast` `GET /admin/hotspots` `POST /admin/recycling/routes/optimize` `PUT /material-prices`

Devices (device key)

`POST /iot/bins/{id}/readings` `POST /iot/hubs/{id}/weighings` `POST /bins/{id}/report-full` `GET /bins`

Selected endpoints read from backend/app/routers. Full contract: docs/api-contract.md.

Appendix A2: the data model in four groups

People

Profile · StaffAccount · Collector · Department · Notification · PushToken

Reports and operations

Incident · Report · IncidentFollower · ServiceRequest · WorkOrder · ActivityLog · Category · BudgetPlan · AISuggestion

CityCycle

RecyclingPickup · Weighing · MaterialPriceRow · SmartBin · BinReading · BinReport · HubWeighing

Safety

IdempotencyKey · OTPChallenge · OTPThrottle

Tables from backend/app/models.py. PostgreSQL with PostGIS on Supabase; Alembic migrations; SQLite for local runs.

Appendix A3: privacy and consent as built today

Staff see the district, not the household address	Verified in tests
Collectors see an approximate pin; exact home cannot be located from the jobs list	Verified in tests
Photo analysis off by default until tested on consented Egyptian samples	To verify
The smart bin has no camera and collects no personal data	Designed
Each device has a key that can be rotated or revoked	Verified in tests
OTP login; demo OTP mode is for demos only	Built

Not a legal compliance claim. A full consent and data-retention policy is still to write.

Appendix A4: risk register

Sensor accuracy in heat and dust	Bench and field tests on a real unit	To verify
GSM or 2G not available	LTE Cat-1 option; test signal at the real location	To verify
Theft and vandalism	Hidden screws, nothing visible, tamper switch	Designed
AI suggests the wrong category	A human confirms; keyword fallback	Built
Demo prices and thresholds	The district sets real prices and thresholds	Planned
No payment integration	Amounts recorded only; integrate a provider later	Planned
Collectors may not adopt it	Pilot with collectors; picture UI; verified pay	To verify
A live deployment is unverified	Runbook in docs/evidence.md before the pitch	To verify

Mitigation tag = status of the mitigation today. Sources: docs/judges-qa.md, docs/evidence.md, docs/field-brief.